Transport in identity-type families is pre- and post-composition

The Univalent Foundations Program

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Editorial note. The counted statement is the author corollary labelled cor:transport-path-prepost (source lines 1875-1886, book number 2.11.2, lemma environment in the source); the proof span 1887-1889 is the author's proof environment, retained line for line.

Editorial note. The author proof is the two-step sentence 'Path induction on p, followed by the unit laws for composition': the induction reduces all three computations to reflexivity and the unit laws (author lemma 2.1.4, proved in place in the source and retained in-corpus as a sibling card) close each case. No step is deferred to the reader or to an exercise.

Editorial scope. One original result of the HoTT book chapter cha:basics, counted once: the pre- and post-computation of transport in identity-type families. Only the counted corollary and its author proof are retained.

Allowed standard prerequisites (editorial).

- A type with an element and paths, and the identity type of that type.
- Transport along a path in a type family, applied to families of identifications.
- Path composition, path inverse, and the unit laws for composition.
- Path induction with its conversion rule.

Lemma 0.0.1. *For any A and $a : A$, with $p : x_1 = x_2$, we have*

$$\begin{aligned}
&\mathsf{transport}^{x \mapsto (a = x)}(p, q) = q \cdot p && \text{for } q : a = x_1,\\
&\mathsf{transport}^{x \mapsto (x = a)}(p, q) = p^{-1} \cdot q && \text{for } q : x_1 = a,\\
&\mathsf{transport}^{x \mapsto (x = x)}(p, q) = p^{-1} \cdot q \cdot p && \text{for } q : x_1 = x_1.
\end{aligned}$$

Proof. Path induction on p, followed by the unit laws for composition. □